



Angelo Nardone

Curriculum Vitae

Personal Information

Date of birth May 17, 1997
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Education

2025–Present **Ph.D. in Computer Science**, *University of Pisa*,
Department of Computer Science,
Supervisor: Paolo Ferragina

2022–2025 **Master's Degree in Computer Science**, *University of Pisa*,
Department of Computer Science,
Curriculum: Artificial Intelligence
Thesis: *Lossless Compression of Source Code using Large Language Models*
Explored novel lossless compression techniques applied to source code in the context of the Software Heritage project, introducing new LLM-based compression pipelines and evaluating 30 different models.
Supervisor: Paolo Ferragina
Final grade: 110/110 cum Laude.

2016–2022 **Bachelor's Degree in Mathematics**, *University of Pisa*,
Department of Mathematics,
Thesis: *Approximate Near Neighbor Searching: Methods for Searching Similar Time Series using the Fréchet Distance*
Supervisor: Roberto Grossi
Final grade: 101/110

Publications and Conferences

2025 **Poster Presentation**, *International Conference on Computational Intelligence Methods for Bioinformatics and Biostatistics (CIBB) 2025, 20th Edition*, Milan, September 10–12, 2025,
Presentation of the short paper "MIEO: encoding clinical data to enhance cardiovascular event prediction", focused on the use of neural models for encoding clinical data to improve cardiovascular event prediction.
Available at: [MIEO: encoding clinical data](#).

Talks and Workshops

- 2026 **Talk**, *Sequences in London 2026*,
City, University of London (Bayes Business School),
London, February 5–6, 2026
Talk title: *Lossless Compression of Source Code using Large Language Models*.
- 2026 **Talk**, *A³Lab Inaugural Workshop: Algorithms, AI & Society*,
Scuola Superiore Sant'Anna,
Pisa, January 21, 2026
Talk title: *Lossless Compression with LLMs*.

Schools

- 2026 **Summer School**, *Agentic AI: algorithms, tools and platforms*,
Lipari School on Computational Complex and Social Systems,
Lipari, June 28 – July 4, 2026
International summer school covering the full lifecycle of autonomous systems, from LLM training and embeddings to vector database architectures, retrieval-augmented generation (RAG), and multi-agent orchestration.
Website: complex26.liparischool.it

Experience

- 2026 **Teaching Tutor (Programming and Algorithms)**, *University of Pisa*,
Pisa,
Awarded a teaching tutoring scholarship in *Programming and Algorithms*.
Activities include tutoring first-year students, guided exercise sessions, and academic support coordinated with the teaching staff.
The activities take place in the second semester (February-June 2026).
- 2025 **Research Fellowship**, *University of Pisa*,
Pisa,
Awarded the research fellowship titled "*Compressione dati tramite Large Language Models (LLM)*", focused on the study and development of compression techniques based on LLMs, with particular application to source code compression.
The fellowship lasts 6 months, starting on July 1, 2025.

Projects

- **Data Compression through LLMs (Research Fellowship Project):** study of lossless data compression on source code using Large Language Models, with the goal of improving compression ratio while keeping throughput as close as possible to traditional methods.
GitHub folder: [Lossless Compression using LLMs](#)
- **Clinical Data Encoding with Autoencoders (Poster Presentation Project):** design of a custom autoencoder for generating clinical embeddings, with automatic handling of missing data and compression of patient health states. This project resulted in the short paper *MIEO: encoding clinical data to enhance cardiovascular event prediction*, presented at CIBB 2025.
GitHub folder: [Clinical Data Encoding](#)

- **Data Mining on U.S. Traffic Accidents:** multivariate analysis on real-world datasets; feature engineering, clustering (K-means, hierarchical, DBSCAN), fatal event classification, and time-based analysis by city.
GitHub folder: [Data Mining](#)
- **Sentiment Analysis on Amazon Reviews:** automatic classification using LLMs and topic modeling to identify recurring negative themes, with applications for user experience and brand reputation.
GitHub folder: [Sentiment Analysis](#)
- **MiniHack Evolution:** training evolutionary agents in NetHack environments (via OpenAI Gym), developing strategies through genetic algorithms in a 15×15 grid-world.
GitHub folder: [MiniHack Evolution](#)

Technical Skills

- Programming Languages
- C/C++: particularly for parallel programming.
 - Python.
 - Java.
 - Matlab.
 - Javascript.

Other Languages Basic knowledge of OCaml, R, Rust and HTML/CSS.

Tools Visual Studio, LaTeX, GitHub.

- Other
- Experience with pandas and torch libraries.
 - Experience working with Large Language Models via HuggingFace.
 - Experience with compression algorithms via libraries and custom implementations.
 - Experience with Linux and macOS systems.

Certifications

2026 **B2.2 English Language Course**, *Linguistic Center, University of Pisa*

2015 **Category B Driver's License**

Wednesday 8th July, 2026

Signature

Angelo Nardone